

Latency Minimization for Uplink RSMA System

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Abstract

Index Terms

I. INTRODUCTION

II. METHODOLOGY

A. Uplink Blocklength Minimization for Reduction of Asynchronicity

For MIMO uplink (UL), the receive signal is written as

$$\mathbf{y} = \sum_{k=1}^K \mathbf{H}_k \mathbf{F}_k \mathbf{s}_k + \mathbf{w} \in \mathbb{C}^{N_R},$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_k}$ denotes the uplink channel between the k th user and the AP, $\mathbf{F}_k \in \mathbb{C}^{N_k \times d_k}$ denotes the precoding matrix for the k th user, d_k denotes the number of spatial streams user k is sending to the AP (and $d_k \leq N_k$), $\mathbf{s}_k \in \mathbb{C}^{d_k}$ denotes the transmitted signal for the k th user, $\mathbf{w} \in \mathbb{C}^{N_R}$ denotes the AWGN, with N_k denoting the number of antennas at the k th user and N_R denotes the number of antennas at the AP. It is assumed that $E[\mathbf{s}_k \mathbf{s}_k^H] = \mathbf{I}_{N_k}$. Let

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{s}_k,$$

so

$$\begin{aligned} E[\|\mathbf{x}_k\|_2^2] &= \mathbf{F}_k E[\mathbf{s}_k \mathbf{s}_k^H] \mathbf{F}_k^H \\ &= \mathbf{F}_k \mathbf{F}_k^H \\ &= \|\mathbf{F}_k\|_F^2 \leq P_k, \end{aligned}$$

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where P_k is the transmit power from the k th user. If SIC is used at the receiver, then the achievable rate at the receiver is

$$R_{sum} = \sum_{k=1}^K \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_k^H \mathbf{H}_k^H \right)$$

Then it seems the latency minimization problem can be formulated as

$$\begin{aligned} \min_{\mathbf{F}_k, B_k, \forall k} \quad & \sum_{k=1}^K \frac{B_k}{R_{sum}} \quad (\text{I think } R_k \\ \text{s.t.} \quad & \|\mathbf{F}_k\|_F^2 \leq P_k, \forall k, \end{aligned} \tag{1}$$

$$B_k \in \{B_{k,\min}, \dots, B_{k,\max}\},$$

where R_k is the Shannon rate for the k th user. (1) makes sense when infinite size blocklength is assumed. However, this assumption is usually incorrect when latency is minimized as the design variable should be the number of blocks (i.e. channel used) for transmission, implying the blocklength is short and finite. Possible extensions for (1) include optimizing not with the sum rate, but using rate such as

1. Max-min rate:

$$\max_{\mathbf{F}_k, \forall k} \min_k R_k.$$

This has a strong fairness guarantee, often leads to equal rates across all users if we can find a feasible solution. This is the worst-case that Carrson was talking about.

2. The proportional fairness:

$$\max_{\mathbf{F}_k, \forall k} \sum_{k=1}^K \log(R_k),$$

which balances the throughput and fairness.

3. Weighted fairness

$$\max_{\mathbf{F}_k, w_k, \forall k} \sum_{k=1}^K w_k \log(R_k),$$

where w_k are weights for QoS differentiation, i.e., higher w_k is, more resource will be allocated the k th user.

B. Uplink Latency Minimization Using RSMA

Using RSMA for uplink transmission can theoretically acheive the optimal rate region according to [1] by splitting the original K input channels to $2K - 1$ inputs (though the K th user message need not be split, but most papers consider splitting the last user message for simplicity). The transmitted message of the k th user $\mathbf{x}_k$ is split into

$$\mathbf{x}_k = \sum_{i=1}^2 F_{ki} \mathbf{s}_{ki},$$

where i denotes the stream index. So the received signal at the AP can be written as

$$\mathbf{y} = \sum_{k=1}^K \sum_{i=1}^2 \mathbf{H}_{ki} \mathbf{F}_{ki} \mathbf{s}_{ki} + \mathbf{w} \in \mathbb{C}^{N_R},$$

where

$$\sum_{i=1}^2 \|\mathbf{F}_{ki}\|_F^2 \leq P_k$$

SIC is assumed to be used to decode each user's signal at the AP, which involves the use of a decoding order π , let the permutation π belong to the set Π defined as the set of all possible decoding order of all $2K$ messages from K users. Let the decoding order of message s_{ki} from the k th user to be π_{ki} , then the achievable rate of decoded message s_{ki} can be given as

missing equation here.

Let $\mathcal{K}$ be the set of user devices, Let $\mathcal{J} = \{1, 2\}$ be the set of containing the number of split streams. Then the rate of the i th stream for user k is

$$R_{ki} = \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_{ki} \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{\{(j \in \mathcal{K}, i \in \mathcal{J}) | \pi_{ji} > \pi_{ki}\}} \mathbf{H}_j \mathbf{F}_{ji} \mathbf{F}_{ji}^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_{ki}^H \mathbf{H}_k^H \right),$$

where $\{(j \in \mathcal{K}, i \in \mathcal{J}) | \pi_{ji} > \pi_{ki}\}$ is the set that represents the messages s_{ji} that are decoded after message s_{ki} . Hence the total rate for user k is

$$R_k = \sum_{i=1}^2 R_{ki}$$

To optimize the rate in uplink RSMA, a joint optimization of both the power management and message decoding order must be done, which results in solving

$$\begin{aligned} & \max_{\mathbf{F}_k, \forall k} \min_{k, \pi} R_k. \\ & \text{s.t.} \quad \sum_{i=1}^2 \|\mathbf{F}_{ki}\|_F^2 \leq P_k \\ & \quad \pi \in \Pi, \|\mathbf{F}_{ki}\|_F^2 \geq 0, \forall k \in \mathcal{K}, j \in \mathcal{J} \end{aligned}$$

This is similar to the optimization problem modeled in [2] where they optimize for the sum rate.

C. Uplink Latency Minimization Using Finite Blocklength Minimization

To consider latency in URRLC systems with finite blocklength, let us model channel rate not in terms of (bits/s) but as (bits/channel use). Let blocklength of the k th user be n_k or the number of channel used per symbol. Let T_s be the time taken to transmit one symbol. Then the latency of the k th user can be expressed as

$$\begin{aligned} \text{Latency}_k &= n_k \times T_s \\ &= \frac{B(\text{bits})}{R(\text{bits/channel use})} \times T_s(\text{sec}) \end{aligned}$$

Thus latency for the k th user is optimized by optimizing for n_k .

According to [3], the **rate under finite blocklength** is expressed as

$$R^*(n, \epsilon) = C - \sqrt{\frac{V}{n}} Q^{-1}(\epsilon) + \mathcal{O}\left(\frac{\log n}{n}\right), \quad (2)$$

where C is Shannon capacity of channel, $Q^{-1}(\cdot)$ is the inverse of the Gaussian Q function, V is the channel dispersion. According to [3], (2) is used to sustain the desired error probability ϵ for a given packet size n , one incurs a penalty on the rate (compared to the channel capacity) that is proportional to $\frac{1}{\sqrt{n}}$.

According to [4], the **rate under finite blocklength in massive MIMO systems** in high SNR regime is approximated as

$$R_k \approx d_k \log(1 + \rho_k) - \sqrt{\frac{d_k}{n_k}} Q^{-1}(\epsilon_k).$$

which is the rate of user k . ρ_k is the SNR of user k and $Q^{-1}(\cdot)$ denotes the inverse of Gaussian Q function. At this point, we are not sure why [4] removed the channel dispersion V , which is part of (2). The inverse of the Q function is

$$Q^{-1}(x) \triangleq \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

Rearranging the equation, when d_k is large enough,

$$n_k \approx \frac{d_k [Q^{-1}(\epsilon_k)]^2}{[d_k \log_2(1 + \rho_k) - R_k]^2} \propto \frac{1}{d_k},$$

This shows that the latency in the time domain can be reduced by increasing the DoFs in the space domain, for the purpose of maintaining a certain level of performance target. According to [4]

$$\begin{aligned} n_k &= B_k / R_k \\ &= \frac{B_k}{d_k \log(1 + \rho_k) - \sqrt{\frac{d_k}{n_k}} Q^{-1}(\epsilon_k)}. \end{aligned} \quad (3)$$

Solving (3) would give another equation that describe n_k if we know the number of bits B_k . We are not sure how this fits in with the other formulas. From (3), it can also be seen there are several ways to decrease n_k , which are

- 1) Increase spatial DoF d_k
- 2) Increase SNR ρ_k
- 3) Increase error probability ϵ_k (reliability)

The **min-max blocklength/Latency** can be expressed as

$$\min_{\forall k} \max_{k, R_k, \epsilon_k} n_k.$$

D. Uplink Latency Minimization with RSMA and Finite Blocklength Minimization

[5] considered minimizing the blocklength in uplink RSMA in a simple 2-user system. However in our problem, we want to improve the latency of the worst-case user, i.e. the user with the worst channel (e.g. worst channel-to-noise (CNR)), we should consider increasing d_k , ρ_k , or ϵ_k of the weakest user through precoder design, while decreasing the same parameters of the stronger users so that all users will experience the same latency, thus reducing asynchronality among the users. Similarly reducing the error probability of the stronger user will be helpful as the data from the stronger users are weighted higher. This helps to reduce the unfairness of asynchronicity and in the context of federated learning, this ensures that the weaker users still have an impact on the training while not weighting the data higher due to higher error rate from these users due to the poor channel. This is something that I do not believe is explored in [5].

E. Problem Formulation of Blocklength Reduction to Reduce Asynchronality

Written on 8-21-2025.

The Shannon rate of the receiver can be defined as,

$$R_k = \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_k^H \mathbf{H}_k^H \right)$$

Defining the latency of the k th user as

$$T_k = \frac{n_k}{f_s},$$

where T_k is the latency, $n_k \in \mathbb{Z}_+$ is the number of channel used and f_s is the symbol rate. Considering error probability ϵ and using the it to calculate latency under the mean no. of retransmissions

$$\begin{aligned} \mathbb{P}(n \text{ transmissions}) &= \epsilon_k^{n-1}(1 - \epsilon_k). \\ \mathbb{E}(\text{no. of transmissions}) &= n_{mean} = \frac{1}{1 - \epsilon_k}. \end{aligned}$$

Sincer a finite blocklength n_k is considered, the upper bound to the rate of system for the k th user is written as [3]

$$\frac{B_k}{n_k} \leq R_k^*(n, \epsilon) = R_k - \sqrt{\frac{V_k}{n_k}} Q^{-1}(\epsilon_k),$$

where B_k is the fixed number of bits that need to sent by the k th user and n_k is the number of blocklength or channel uses or symbols for the k th user. Thus $R_k^*(n, \epsilon)$ can be defined in the unit of (bits/channel use). Then the weighted latency minimization problem can be formulated as

$$\begin{aligned} \min_{\mathbf{F}_k, n_k, \epsilon_k} \quad & \sum_{k \in \mathcal{K}} W_k \frac{n_k}{f_s} \\ \text{s.t.} \quad & \frac{B_k}{n_k} \leq R_k(\mathbf{F}_k) - \sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}} Q^{-1}(\epsilon_k), \quad \forall k \in \mathcal{K} \\ & n_k \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K} \\ & \|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}. \end{aligned} \tag{4}$$

A more complete formulation by considering user specific ϵ_k and retransmission affected latency can be considered as

$$\begin{aligned} \min_{\mathbf{F}_k, n_k, \epsilon_k} \quad & \sum_{k \in \mathcal{K}} W_k \frac{n_k}{f_s} \frac{1}{1 - \epsilon_k} \\ \text{s.t.} \quad & \frac{B_k}{n_k} \leq (1 - \epsilon_k) \left[R_k(\mathbf{F}_k) - \sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}} Q^{-1}(\epsilon_k) \right], \quad \forall k \in \mathcal{K}, \\ & n_k \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K}, \\ & \epsilon_k \in [0, \epsilon_0], \quad \forall k \in \mathcal{K}, \\ & \|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}. \end{aligned} \tag{5}$$

F. Strategies to solve the Problem Formulation

Solution for (10) can be attained via alternating optimization. First, by fixing $\mathbf{F}_k$, n_k can be found via exhaustive search by searching the interval $[n_{\min}, n_{\max}]$ for $k \in \mathcal{K}$. Then after a

suboptimal (because $n_k, \forall k$ is found based on a suboptimal $\mathbf{F}_k$), then fixing $\mathbf{n}_k$, we should find $\mathbf{F}_k$, for $k \in \mathcal{K}$, by solving

$$\begin{aligned} \max_{\mathbf{F}_k, k \in \mathcal{K}} R_k(\mathbf{F}_k) - \sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}} Q^{-1}(\epsilon_k), \quad \forall k \in \mathcal{K} \\ \|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}. \end{aligned} \quad (6)$$

This implies if the problem needs to be convex, then $R_k(\mathbf{F}_k)$ needs to be concave (where the maximization of $R_k(\mathbf{F}_k)$ can be done by reformulating the problem into minimization of the WMMSE) and $\sqrt{\frac{V_k(\mathbf{F}_k)}{n_k}}$ needs to be convex. Since square root is a monotonically nondecreasing concave function, the square root can be removed and $V_k(\mathbf{F}_k)$ can be approximated, e.g. using its first-order Taylor series expansion, so that (6) can be convexified as

$$\begin{aligned} \max_{\mathbf{F}_k, k \in \mathcal{K}} W_k(\mathbf{F}_k) - \left(V_k(\mathbf{F}_k^{(p)}) + \nabla V_k(\mathbf{F}_k^{(p)})^H (\mathbf{F}_k - \mathbf{F}_k^{(p)}) \right), \quad \forall k \in \mathcal{K} \\ \|\mathbf{F}_k\|_F^2 \leq P_k, \quad \forall k \in \mathcal{K}, \end{aligned} \quad (7)$$

where $\mathbf{F}_k^{(p)}$ is the precoder at the p th iteration of the successive convex optimization (SCA) (or the WMMSE update since the SCA needs to be applied in conjunction with the WMMSE iteration).

Since the rate of the k th user is defined as

$$R_k = \log \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^H \mathbf{H}_k^H \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \right),$$

let $\mathbf{H}_k \in \mathbb{C}^{N_r \times N_k}$ with N_k denoting the number of transmit antennas at user k . This implies $\mathbf{F}_k \in \mathbb{C}^{N_k \times d_k}$, where $d_k \leq N_k$ denotes the number of spatial streams from the k th user. Then it has been shown in [6] that the weighted sum rate

$$\sum_{k=1}^K w_k R_k, \quad s.t. \|\mathbf{F}_k\|_F^2 \leq P_k$$

can be maximized by solving the WMMSE problem. Define the k th receiver at the AP as $\mathbf{U}_k \in \mathbb{C}^{N_R \times d_k}$. Also define the weight matrix $\mathbf{W}_k \in \mathbb{C}^{d_k \times d_k}$ where $\mathbf{W}_k \succ \mathbf{0}_{d_k \times d_k}$ (which will be equal to $\mathbf{E}_k^{-1}$). Further define the MSE matrix

$$\mathbf{E}_k \triangleq \mathbf{I}_{d_k} - \mathbf{U}_k^H \mathbf{H}_k \mathbf{F}_k - \mathbf{F}_k^H \mathbf{H}_k^H \mathbf{U}_k + \mathbf{U}_k^H \left(\sigma_w^2 \mathbf{I}_{N_r} + \sum_{j=1}^K \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right) \mathbf{U}_k \in \mathbb{C}^{d_k \times d_k}.$$

Then the per user matrix quadratic transform identity implies

$$\begin{aligned} w_k \log \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^H \mathbf{H}_k^H \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \right) \\ = \max_{\mathbf{U}_k, \mathbf{W}_k \succ \mathbf{0}_{d_k \times d_k}} w_k (\log \det \mathbf{W}_k - \text{tr}(\mathbf{W}_k \mathbf{E}_k)) + \text{constant}. \end{aligned}$$

Then the original weighted sum rate problem $\max_{\mathbf{F}_k, k \in \mathcal{K}} w_k R_k$, subject to $\|\mathbf{F}_k\|_F^2 \leq P_k$ problem is equivalent to

$$\begin{aligned} \max_{\mathbf{F}_k, \mathbf{W}_k, \mathbf{U}_k, k \in \mathcal{K}} \quad & w_k (\log \det \mathbf{W}_k - \text{tr}(\mathbf{W}_k \mathbf{E}_k)) \\ \text{s.t.} \quad & \|\mathbf{F}_k\|_F^2 \leq P_k. \end{aligned} \quad (8)$$

Let

$$\Sigma \triangleq \sigma_w^2 \mathbf{I}_{N_R} + \sum_{j=1}^K \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H.$$

Then the following WMMSE algorithm can be used to find the solution for (8).

1. MMSE receiver update

$$\mathbf{U}_k = \Sigma^{-1} \mathbf{H}_k \mathbf{F}_k.$$

2. Weight matrix computation (inverse MSE)

$$\mathbf{W}_k = \mathbf{E}_k^{-1}$$

3. Precoder computation (quadratic KKT with power constraint) Let

$$\mathbf{C}_k \triangleq \mathbf{H}_k^H \mathbf{U}_k \mathbf{W}_k \mathbf{U}_k^H \mathbf{H}_k \in \mathbb{C}^{N_k \times N_k}$$

$$\mathbf{D}_k \triangleq \mathbf{H}_k^H \mathbf{U}_k \mathbf{W}_k \in \mathbb{C}^{N_k \times d_k}.$$

Then the precoder with per-user power constraint is computed as

$$\mathbf{F}_k(\lambda_k) = (\mathbf{w}_k \mathbf{C}_k + \lambda_k \mathbf{I}_{N_k})^{-1} (\mathbf{w}_k \mathbf{D}_k) \in \mathbb{C}^{N_k \times d_k},$$

where $\lambda_k \geq 0$ is chosen by the bisection method to satisfy $\text{tr}(\mathbf{F}_k \mathbf{F}_k^H) = P_k$ if needed, otherwise $\lambda_k = 0$ and $\text{tr}(\mathbf{F}_k \mathbf{F}_k^H) < P_k$. That is, this corresponds to the complementary slackness condition in the KKT conditions.

Note that it has been reported that by initializing $\mathbf{F}_k$ using MRT precoder speeds up the convergence of the WMMSE algorithm. Note also that the sum rate is maximized by maximizing the per-user rate R_k . Note that we may be able to set the weight w_k as the inverse of the CNR so that the rate of the worst-case (i.e. worst CNR) will be optimized the most. Algorithm 1 summarizes the steps.

Algorithm 1 Proposed latency minimization algorithm in the uplink via precoder.

Ensure: $n_k, \mathbf{F}_k^*, \forall k$

Require: Set $i = 0$, choose ε (e.g. small integer number), choose $\{\mathbf{F}_k^{(0)}, \forall k\}$ as the MRT precoder, $w_k = \text{inverse of the } \text{CNR}_k$

```

1: repeat
2:   Exhaustively find  $n_k, \forall k$  using  $\mathbf{F}_k^{(i)}$ 
3:   Denote (sub-)optimal blocklength as  $n_k^*$ 
4:    $n_k^{(i)} = n_k^*, \forall k$ 
5:   for each user  $k = 1, \dots, K$  do
6:     Use the WMMSE algorithm and convex approximation of  $V_k(\mathbf{F}_k)$  to find the
       precoder  $\mathbf{F}_k$  using  $n_k^{(i)}$ 
7:     Denote (sub-)optimal precoder as  $\mathbf{F}_k^*$ 
8:   end for
9:    $i = i + 1$ 
10:   $\mathbf{F}_k^{(i)} = \mathbf{F}_k^*, \forall k$ 
11: until  $\left| \sum_{k \in \mathcal{K}} n_k^{(i)} - \sum_{k \in \mathcal{K}} n_k^{(i-1)} \right| \leq \varepsilon$ 

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G. Capacity and Channel Dispersion

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The Capacity and Channel Variance of Coherent Rayleigh Fading Channel is discussed in [7].

However [7] makes the following assumptions are made,

- 1) The channel is isotropic.
- 2) The noise has unit variance.
- 3) Considers SU-MIMO.

The Capacity of the channel of user- k is defined as,

$$\begin{aligned}
 C(P) &= \frac{1}{2} \mathbb{E} \left[\log \det \left(I_{n_r} + \frac{P}{n_t} \mathbf{H} \mathbf{H}^\dagger \right) \right] \\
 &= \sum_{i=1}^{n_{\min}} \mathbb{E} \left[C_{\text{AWGN}} \left(\frac{P}{n_t} \lambda_i^2 \right) \right],
 \end{aligned}$$

where $C_{\text{AWGN}}(P) = \frac{1}{2} \log(1 + P)$ is the AWGN capacity with unit noise variance, $n_{\min} = \min(n_r, n_t)$, and $\lambda_i^2, i \in \{1, \dots, n_{\min}\}$ are the eigenvalues of $\mathbf{H} \mathbf{H}^\dagger$.

Channel Variance

Assume that $\mathbb{P}[\text{rank}\mathbf{H} > 1] > 0$, then $V(P) = V_{iid}(P)$,

$$\begin{aligned} V_{iid}(P) &= T \text{Var} \left(\sum_{i=1}^{n_{\min}} C_{\text{AWGN}} \left(\frac{P}{n_t} \lambda_i^2 \right) \right) \\ &\quad + \sum_{i=1}^{n_{\min}} \mathbb{E} \left[V_{\text{AWGN}} \left(\frac{P}{n_t} \lambda_i^2 \right) \right] \\ &\quad + \left(\frac{P}{n_t} \right)^2 \left(\eta_1 - \frac{\eta_2}{n_t} \right) \end{aligned}$$

where λ_i^2 , $i \in \{1, \dots, n_{\min}\}$ are the eigenvalues of $\mathbf{H}\mathbf{H}^H$.

$$V_{\text{AWGN}}(P) = \frac{(\log e)^2}{2} \left(1 - \frac{1}{(1+P)^2} \right)$$

$$\begin{aligned} c(\sigma) &\triangleq \frac{\sigma}{1 + \frac{P}{n_t} \sigma} \\ \eta_1 &\triangleq \frac{(\log e)^2}{2} \sum_{i=1}^{n_{\min}} \mathbb{E} [c^2(\lambda_i^2)] \\ \eta_2 &\triangleq \frac{(\log e)^2}{2} \left(\sum_{i=1}^{n_{\min}} \mathbb{E} [c(\lambda_i^2)] \right)^2 \end{aligned}$$

Obtaining a closed form

Under high SNR $P \rightarrow \infty$. Let $n_{\min} = \min(n_t, n_r)$, $\alpha = P/n_t$ and λ_i the eigenvalues of $\mathbf{H}\mathbf{H}^H$.

Reducing C_{AWGN} ,

Capacity $C_{\text{AWGN}}(x) = \log_e 1 + x$ and in $V_{iid}(P)$,

$$T \text{Var} \left[\sum_{i=1}^{n_{\min}} C_{\text{AWGN}} \left(\frac{P}{n_t} \lambda_i^2 \right) \right] = T \text{Var} \left[\sum_{i=1}^{n_{\min}} \ln(1 + \alpha \lambda_i^2) \right], \quad \alpha = \frac{P}{n_t}.$$

$$\text{Var} \left[\sum_{i=1}^{n_{\min}} \ln(1 + \alpha \lambda_i^2) \right] \sim \text{Var} \left[\sum_{i=1}^{n_{\min}} \ln \lambda_i^2 \right]$$

Reducing V_{AWGN} ,

$$V_{\text{AWGN}}(P) = \frac{(\log e)^2}{2} \left(1 - \frac{1}{(1+P)^2} \right)$$

$$\begin{aligned}
c(\sigma) &\triangleq \frac{\sigma}{1 + \frac{P}{n_t} \sigma}, \\
\eta_1 &\triangleq \frac{(\log e)^2}{2} \sum_{i=1}^{n_{\min}} \mathbb{E}[c^2(\lambda_i^2)], \\
\eta_2 &\triangleq \frac{(\log e)^2}{2} \left(\sum_{i=1}^{n_{\min}} \mathbb{E}[c(\lambda_i^2)] \right)^2. \\
c(\sigma) &\triangleq \frac{\sigma}{1 + \frac{P}{n_t} \sigma}, \quad \alpha \triangleq \frac{P}{n_t}.
\end{aligned}$$

Reducing $\left(\frac{P}{n_t}\right)^2(\eta_1 - \frac{\eta_2}{n_t})$,

Step 1: High-SNR expansion of $c(\sigma)$

$$c(\sigma) = \frac{\sigma}{1 + \alpha\sigma} = \frac{1}{\alpha} \cdot \frac{1}{1 + \frac{1}{\alpha\sigma}}.$$

Using the series $\frac{1}{1+x} = 1 - x + x^2 + \mathcal{O}(x^3)$ with $x = \frac{1}{\alpha\sigma}$:

$$c(\sigma) = \frac{1}{\alpha} - \frac{1}{\alpha^2\sigma} + \frac{1}{\alpha^3\sigma^2} + \mathcal{O}\left(\frac{1}{\alpha^4}\right).$$

—

Step 2: Expansion of η_1

$$c(\sigma)^2 = \frac{1}{\alpha^2} - \frac{2}{\alpha^3\sigma} + \mathcal{O}\left(\frac{1}{\alpha^4}\right).$$

$$\eta_1 = \frac{\log^2 e}{2} \sum_{i=1}^{n_{\min}} \mathbb{E}[c(\lambda_i^2)^2] = \frac{\log^2 e}{2} \left(\frac{n_{\min}}{\alpha^2} - \frac{2}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\frac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right).$$

—

Step 3: Expansion of η_2

$$\mathbb{E}[c(\lambda_i^2)] = \frac{1}{\alpha} - \frac{1}{\alpha^2} \mathbb{E}\left[\frac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^3}\right).$$

$$\sum_{i=1}^{n_{\min}} \mathbb{E}[c(\lambda_i^2)] = \frac{n_{\min}}{\alpha} - \frac{1}{\alpha^2} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\frac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^3}\right).$$

$$\left(\sum_{i=1}^{n_{\min}} \mathbb{E}[c(\lambda_i^2)] \right)^2 = \frac{n_{\min}^2}{\alpha^2} - \frac{2n_{\min}}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\frac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^4}\right).$$

$$\eta_2 = \frac{\log^2 e}{2} \left(\frac{n_{\min}^2}{\alpha^2} - \frac{2n_{\min}}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\frac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right).$$

Step 4: Reducing $(\frac{P}{n_t})^2(\eta_1 - \frac{\eta_2}{n_t})$ $(\frac{P}{n_t})^2(\eta_1 - \frac{\eta_2}{n_t}) = \alpha^2(\eta_1 - \frac{\eta_2}{n_t})$

$$\eta_1 - \frac{\eta_2}{n_t} = \frac{\log^2 e}{2} \left(\frac{n_{\min} - \frac{n_{\min}^2}{n_t}}{\alpha^2} + \frac{2(\frac{n_{\min}}{n_t} - 1)}{\alpha^3} \sum_{i=1}^{n_{\min}} \mathbb{E} \left[\frac{1}{\lambda_i^2} \right] + \mathcal{O} \left(\frac{1}{\alpha^4} \right) \right).$$

$$\left(\frac{P}{n_t} \right)^2 n_t (\eta_1 - \eta_2) = \alpha^2 (\eta_1 - \eta_2 / n_t) = \frac{\log^2 e}{2} \left(n_{\min} - \frac{n_{\min}^2}{n_t} + \frac{2(\frac{n_{\min}}{n_t} - 1)}{\alpha} \sum_{i=1}^{n_{\min}} \mathbb{E} \left[\frac{1}{\lambda_i^2} \right] + \mathcal{O} \left(\frac{1}{\alpha^2} \right) \right).$$

If we assume that $\alpha \approx \infty$,

$$\left(\frac{P}{n_t} \right)^2 (\eta_1 - \frac{\eta_2}{n_t}) = \frac{\log^2 e}{2} \left(n_{\min} - \frac{n_{\min}^2}{n_t} + \mathcal{O} \left(\frac{1}{\alpha^2} \right) \right)$$

Final Value of $(\frac{P}{n_t})^2(\eta_1 - \frac{\eta_2}{n_t})$

Having $(\frac{P}{n_t})^2(\eta_1 - \frac{\eta_2}{n_t})$ to be independant of precoder $\mathbf{F}_k$ by eliminating $\alpha = \frac{P}{n_t}$,

$$\left(\frac{P}{n_t} \right)^2 (\eta_1 - \frac{\eta_2}{n_t}) = \frac{\log^2 e}{2} \left(n_{\min} - \frac{n_{\min}^2}{n_t} + \mathcal{O} \left(\frac{1}{\alpha^2} \right) \right) \quad \text{as } P \rightarrow \infty.$$

Having $(\frac{P}{n_t})^2(\eta_1 - \frac{\eta_2}{n_t})$ to be dependant on precoder $\mathbf{F}_k$ by including $\alpha = \frac{P}{n_t}$,

$$\left(\frac{P}{n_t} \right)^2 (\eta_1 - \eta_2) = \alpha^2 (\eta_1 - \eta_2 / n_t) = \frac{\log^2 e}{2} \left(n_{\min} - \frac{n_{\min}^2}{n_t} + \frac{2(\frac{n_{\min}}{n_t} - 1)}{\alpha} \sum_{i=1}^{n_{\min}} \mathbb{E} \left[\frac{1}{\lambda_i^2} \right] + \mathcal{O} \left(\frac{1}{\alpha^2} \right) \right).$$

Reducing $T \text{Var}[\sum_{i=1}^{n_{\min}} \ln \lambda_i^2]$,

We can further expand $T \text{Var}[\sum_{i=1}^{n_{\min}} \ln \lambda_i^2]$ with the trigamma function $\psi_1(x)$

$$\text{Var} \left[\sum_{i=1}^{n_{\min}} \ln \lambda_i^2 \right] = \sum_{i=1}^{n_{\min}} \psi_1(n_{\max} - i + 1),$$

$$n_{\min} = \min(n_t, n_r), \quad n_{\max} = \max(n_t, n_r),$$

$$\alpha = \frac{P}{n_t}$$

where $\psi_1(x)$ is,

$$\psi_1(x) = \sum_{k=0}^{\infty} \frac{1}{(x+k)^2}.$$

In the assumption of very large x ,

$$\psi_1(x) = \frac{1}{x} + \frac{1}{2x^2} + \frac{1}{6x^3} - \frac{1}{30x^5} + \frac{1}{42x^7} - \dots$$

Thus when we consider $\sum_{i=1}^{n_{\min}} \psi_1(n_{\max} - i + 1)$ to obtain a closed form for the trigamma function, we need to assume a very large $n_{\max} - i + 1$, thus if $n_{\max} = \max(n_r, n_t) \rightarrow \infty$,

$$\begin{aligned}\psi_1(n_{\max} - i + 1) &= \frac{1}{n_{\max} - i + 1} + \frac{1}{2(n_{\max} - i + 1)^2} \\ &\quad + \frac{1}{6(n_{\max} - i + 1)^3} - \frac{1}{30(n_{\max} - i + 1)^5} + \dots\end{aligned}$$

or using the Bernoulli numbers B ,

$$\psi_1(n_{\max} - i + 1) = \sum_{k=0}^{\infty} \frac{B_{2k}}{(n_{\max} - i + 1)^{2k+1}}$$

Thus we can reduce $\text{Var}[\sum_{i=1}^{n_{\min}} \ln \lambda_i^2]$ to,

$$\text{Var} \left[\sum_{i=1}^{n_{\min}} \ln \lambda_i^2 \right] \approx \sum_{i=1}^{n_{\min}} \psi_1(n_{\max} - i + 1) = \sum_{i=1}^{n_{\min}} \left[\frac{1}{n_{\max} - i + 1} + \frac{1}{2(n_{\max} - i + 1)^2} + \dots \right]$$

Final closed form for Channel Variance

Thus the final $V(P)$ under the assumption of, $P \rightarrow \infty$ and $n_{\max} \rightarrow \infty$ is,

Case 1: when $V(P)$ is independent on precoder $\mathbf{F}_k$,

$$\begin{aligned}V(P) &= V_{\text{iid}}(P) \approx \sum_{i=1}^{n_{\min}} \left[\frac{1}{n_{\max} - i + 1} + \frac{1}{2(n_{\max} - i + 1)^2} + \dots \right] \\ &\quad + n_{\min} (\ln e)^2 \\ &\quad + \frac{\log^2 e}{2} \left(n_{\min} - \frac{n_{\min}^2}{n_t} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \right) \quad \alpha = \frac{P}{n_t}.\end{aligned}$$

Case 2: when $V(P)$ is dependent on precoder $\mathbf{F}_k$,

$$\begin{aligned}V(P) &= V_{\text{iid}}(P) \approx \sum_{i=1}^{n_{\min}} \left[\frac{1}{n_{\max} - i + 1} + \frac{1}{2(n_{\max} - i + 1)^2} + \dots \right] \\ &\quad + n_{\min} (\ln e)^2 \\ &\quad + \frac{\log^2 e}{2} \left(n_{\min} - \frac{n_{\min}^2}{n_t} + \frac{2\left(\frac{n_{\min}}{n_t}\right) - 1}{\alpha} \sum_{i=1}^{n_{\min}} \mathbb{E}\left[\frac{1}{\lambda_i^2}\right] + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \right). \quad \alpha = \frac{P}{n_t}.\end{aligned}$$

H. Revised Uplink Blocklength Minimization for Reduction of Asynchronicity in Rayleigh Fading Channel

Edited on 29-09-25

In the classical infinite blocklength regime, the achievable rate of a user in a MIMO uplink channel is characterized by the Shannon capacity expression

$$R_k = \log \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \mathbf{F}_k^H \mathbf{H}_k^H \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \right),$$

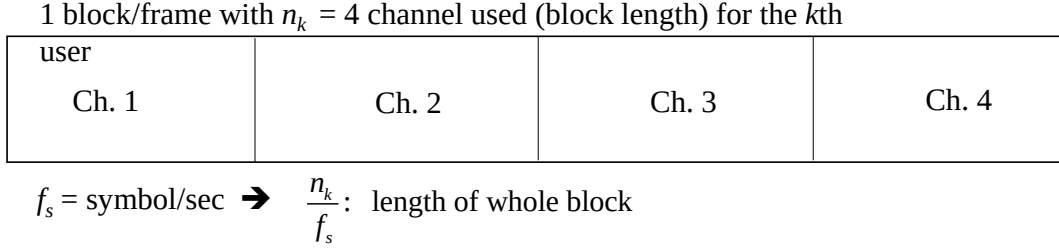


Fig. 1: Illustration of blocklength (or channel used) in a block/frame of bits. The figure shows that $n_k = 4$, so $\frac{n_k}{f_s}$ denotes the length of the whole block/frame in time (sec).

where $\mathbf{H}_k$ denotes the channel matrix of the k th user, $\mathbf{F}_k$ is the precoding matrix and σ_w^2 the noise variance. This formulation assumes that the blocklength, i.e., the number of channel uses for transmitting a codeword, tends to infinity, which enables arbitrarily small error probability through sufficiently long channel codes. This is illustrated in Fig. 1

In reality most low latency application operate in the finite blocklength regime, where the number of transmitted symbols n is limited. In this case in a Gaussian channel, the received signal with blocklength n can be written as

$$\mathbf{Y} = \sum_{k=1}^K \mathbf{H}_k \mathbf{F}_k \mathbf{S}_k + \mathbf{W}$$

where the received signal $\mathbf{Y} \in \mathbb{C}^{N_R \times n}$, the transmitted signal $\mathbf{S}_k \in \mathbb{C}^{d_k \times n}$, the Gaussian channel matrix $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_k}$, the precoder matrix $\mathbf{F}_k \in \mathbb{C}^{N_k \times d_k}$, the noise matrix $\mathbf{W} \in \mathbb{C}^{N_R \times n}$ and $d_k \leq N_k$. N_k denotes the number of transmitted antenna at the k th user.

However this assumes a fixed Gaussian channel, a time-varying Rayleigh fading will be considered herein. Let T denote the coherence time of the channel, i.e. the number of channel uses over which the channel remains constant. Then the blocklength (total channel uses) of a codeword can be written as **THIS EQUATION STILL DOESN'T MAKE SENSE TO ME.** If T is the channel used/coherent interval, then it should always be 1 as 1 channel used occupies 1 coherent interval

$$n = L \times T$$

where T is the number of channel uses over which the channel remains constant and L is the number of coherent channels within one blocklength.

By considering the coherent intervals, the received signal can be written as
From now on, don't use "l", because that will be confused as "1" (one). Use ℓ instead

$$\mathbf{Y} = \sum_{k=1}^K \sum_{\ell=1}^L \mathbf{H}_{k,\ell} \mathbf{F}_{k,\ell} \mathbf{S}_{k,\ell} + \mathbf{W} \in \mathbb{C}^{L \times N_r \times T}$$

where $\mathbf{Y} \in \mathbb{C}^{L \times N_r \times T}$, $\mathbf{H}_k \in \mathbb{C}^{L \times N_r \times N_k}$, $\mathbf{F}_k \in \mathbb{C}^{L \times N_k \times d_k}$, $\mathbf{S}_k \in \mathbb{C}^{L \times d_k \times T}$

~~(i think it can also be written as $\mathbf{Y} \in \mathbb{C}^{N_r \times L \times T}$, $\mathbf{H}_k \in \mathbb{C}^{N_r \times L \times N_k}$, $\mathbf{F}_k \in \mathbb{C}^{N_k \times L \times d_k}$, $\mathbf{S}_k \in \mathbb{C}^{d_k \times L \times T}$)~~

I. Problem Formulation with Rayleigh Fading Channel

Let B_k be number of bits sent by the k th user, let n_k be the blocklength of user k , let $\mathbf{F}_k$ be the precoder of user- k . Let blocklength of user k be split in coherent intervals as **SAME HERE. THIS EQUATION STILL DOESN'T MAKE SENSE TO ME.** If T_K is the channel used/coherent interval, then it should always be 1 as 1 channel used occupies 1 coherent interval

$$n_k = L_k \times T_K$$

where L_k is the number of coherent intervals within the blocklength of user k (cf. Fig. 1, this can mean bv) and T_k is the number of channel uses in one coherent interval of user k .

A weighted latency minimization problem can be defined as

$$\begin{aligned} \min_{\mathbf{F}_{k,\ell}, n_k, \epsilon_k} \quad & \sum_{k \in \mathcal{K}} w_k \frac{n_k}{f_s} \\ \text{s.t.} \quad & \frac{B_k}{n_k} \leq \max_{\mathbf{F}_{k,\ell}} \frac{1}{L_k} \sum_{\ell=1}^{L_k} R^*(T_k, \mathbf{F}_{k,\ell}), \quad \forall k \in \mathcal{K} \\ & n_k \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K} \\ & \|\mathbf{F}_{k,\ell}\|_F^2 \leq P_{k,\ell}, \quad \forall \ell \in [0, L_k], \forall k \in \mathcal{K}, \end{aligned} \quad (9)$$

where the finite blocklength rate for coherent interval ℓ of user k is defined as,

$$R^*(T_k, \mathbf{F}_{k,\ell}) = C_k(\mathbf{F}_{k,\ell}) - \sqrt{\frac{V_k(\mathbf{F}_{k,\ell})}{T_{k,\ell}}} Q^{-1}(\epsilon_k), \quad \forall \ell \in [0, L_k] \forall k \in \mathcal{K}$$

The channel variance of the k th user in coherence interval ℓ is defined as,

$$\begin{aligned} V_k(P_{k,\ell}) \approx & \sum_{i=1}^{N_{k,\min}} \left[\frac{1}{N_{k,\max} - i + 1} + \frac{1}{2(N_{k,\max} - i + 1)^2} + \dots \right] \\ & + N_{k,\min} (\ln e)^2 \\ & + \frac{(\ln e)^2}{2} \left(N_{k,\min} - \frac{N_{k,\min}^2}{N_k} + \frac{2 \left(\frac{N_{k,\min}}{N_k} - 1 \right)}{\alpha_{k,\ell}} \sum_{i=1}^{N_{k,\min}} \mathbb{E} \left[\frac{1}{\lambda_{k,\ell,i}^2} \right] + \mathcal{O} \left(\frac{1}{\alpha_{k,\ell}^2} \right) \right), \end{aligned}$$

where $\alpha_{k,\ell} = \frac{P_{k,\ell}}{N_k}$ is the SNR and $P_{k,\ell}$ is the transmit power. However since we are assuming an isotropic Rayleigh fading channel it is also equal to the total transmit power thus, $P_{k,\ell} = \|\mathbf{F}_{k,\ell}\|_F^2$.

J. Solution in case of Rayleigh Fading Channel

$$\begin{aligned}
& \min_{\mathbf{F}_{k,\ell}, n_k, \epsilon_k} \sum_{k \in K} w_k \frac{n_k}{f_s} \\
& \text{s.t. } \frac{B_k}{n_k} \leq \max_{\mathbf{F}_{k,\ell}} \frac{1}{L_k} \sum_{\ell=1}^{L_k} R^*(T_k, \mathbf{F}_{k,\ell}), \quad \forall k \in \mathcal{K} \\
& n_k \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K} \\
& \|\mathbf{F}_{k,\ell}\|_F^2 \leq P_{k,\ell}, \quad \forall \ell \in [0, L_k], \forall k \in \mathcal{K}.
\end{aligned} \tag{10}$$

(10) can be solved by first fixing $\mathbf{F}_{k,\ell}$ and finding n_k via exhaustive search between $[n_{\min}, n_{\max}]$.

This is followed by fixing n_k to the value that was found in the previous step and maximizing $R^*(T_k, \mathbf{F}_{k,\ell})$ to reduce the latency by increasing the upper bound in the 1st constraint of (10).

To convert $R^*(T_k, \mathbf{F}_{k,\ell})$ into a concave function, the square root in $\sqrt{\frac{V(\mathbf{F}_{k,\ell})}{T_k}} Q^{-1}(\epsilon_k)$ is removed, followed by variable substitution by solving $\mathbf{Q}_{k,\ell} \triangleq \mathbf{F}_{k,\ell} \mathbf{F}_{k,\ell}^H$ instead of $\mathbf{F}_{k,\ell}$. This makes $R^*(T_k, \mathbf{Q}_{k,\ell})$ a concave function. Then $R^*(T_k, \mathbf{Q}_{k,\ell})$ is maximized as

$$\max_{\mathbf{F}_{k,\ell} \in \mathcal{C}} C_k(\mathbf{F}_{k,\ell}) - V(\mathbf{F}_{k,\ell}) = \max_{\mathbf{F}_{k,\ell} \in \mathcal{C}} \log \det \left(\sigma_w^{-2} \mathbf{H}_{k,\ell} \mathbf{F}_{k,\ell} \mathbf{F}_{k,\ell}^H \mathbf{H}_{k,\ell}^H \right) - \frac{V(\mathbf{F}_{k,\ell})}{T_k} Q^{-1}(\epsilon_k), - \frac{V(\mathbf{Q}_{k,\ell})}{T_k} Q^{-1}(\epsilon_k) \tag{11}$$

where $\mathcal{C} = \left\{ \mathbf{F}_{k,\ell} \left| \frac{B_k}{n_k} \leq \max_{\mathbf{F}_{k,\ell}} \frac{1}{L_k} \sum_{\ell=1}^{L_k} R^*(T_k, \mathbf{F}_{k,\ell}), \|\mathbf{F}_{k,\ell}\|_F^2 \leq P_{k,\ell}, \forall \ell \in [0, L_k], \text{ for } k = 1, \dots, K, \right. \right\}$.

The solution for both n_k and $\mathbf{Q}_{k,\ell}$ is obtained through alternating optimization by solving (10) and (11). Once the optimal $\mathbf{Q}_{k,\ell}$ is obtained, $\mathbf{F}_{k,\ell}$ can be found via a randomization procedure that satisfies the transmit power constraint $\|\mathbf{F}_{k,\ell}\|_F^2 \leq P_{k,\ell}, \forall \ell \in [0, L_k], \forall k \in \mathcal{K}$ as outlined in [8].

Since it appears to be difficult to convexify (11) due to the complex nature of $V(\mathbf{F}_{k,\ell})$, (10) can be solved using a neural network. Finding n_k is the same as before, i.e. using exhaustive search. However, solving for (11) will be different. Since the problem is a constrained problem, its Lagrangian can be written as

$$\mathcal{L}(\mathbf{F}_{k,\ell}, \lambda_1, \lambda_2) = R^*(T_k, \mathbf{F}_{k,\ell}) + \lambda_1 \left(R^*(T_k, \mathbf{F}_{k,\ell}) - \frac{B_k}{n_k} \right) + \lambda_2 (P_{k,\ell} - \text{tr}(\mathbf{F}_{k,\ell} \mathbf{F}_{k,\ell}^H)).$$

(11) can be solved via the primal-dual algorithm by breaking it into the following three

problems

$$\mathbf{F}_{k,\ell}^{(i+1)} = \arg \max_{\mathbf{F}_{k,\ell}^{(i)}} \mathcal{L}(\mathbf{F}_{k,\ell}^{(i)}, \lambda_1^{(i)}, \lambda_2^{(i)}) \quad (12a)$$

$$\begin{aligned} \lambda_1^{i+1} &= \arg \min_{\lambda_1} \mathcal{L}(\mathbf{F}_{k,\ell}^{(i+1)}, \lambda_1, \lambda_2^{(i)}) \\ &= \left[\lambda_1^{(i)} - \eta_{\lambda_1}^{(i)} \left(R^*(T_k, \mathbf{F}_{k,\ell}^{(i+1)}) - \frac{B_k}{n_k^{(i+1)}} \right) \right]_+ \end{aligned} \quad (12b)$$

$$\begin{aligned} \lambda_2^{(i+1)} &= \arg \min_{\lambda_2} \mathcal{L}(\mathbf{F}_{k,\ell}^{(i+1)}, \lambda_1^{(i+1)}, \lambda_2) \\ &= \left[\lambda_2^{(i)} - \eta_{\lambda_2}^{(i)} \left(P_{k,\ell} - \text{tr} \left(\mathbf{F}_{k,\ell}^{(i+1)} \mathbf{F}_{k,\ell}^{(i+1)H} \right) \right) \right]_+. \end{aligned} \quad (12c)$$

According to the complementary slackness condition,

$$\begin{aligned} \lambda_1 \left(R^*(T_k, \mathbf{F}_{k,\ell}) - \frac{B_k}{n_k} \right) &= 0 \\ \lambda_2 (P_{k,\ell} - \text{tr}(\mathbf{F}_{k,\ell} \mathbf{F}_{k,\ell}^H)) &= 0, \end{aligned}$$

which implies that either $\lambda_1 = 0$ or $R^*(T_k, \mathbf{F}_{k,\ell}) - \frac{B_k}{n_k} = 0$ and $\lambda_2 = 0$ or $P_{k,\ell} - \text{tr}(\mathbf{F}_{k,\ell} \mathbf{F}_{k,\ell}^H) = 0$ when the primal-dual algorithm converges. The iteration should be terminated using the above complementary slackness conditions, as well as the other relevant KKT conditions.

(12a) can be solved by parameterizing the solution using a neural network with parameters θ so that it can be found using gradient ascent

$$\theta_{k,\ell}^{(i+1)} = \theta_{k,\ell}^{(i)} + \eta_{\theta_{k,\ell}}^{(i)} \nabla_{\theta_{k,\ell}^{(i)}} \mathcal{L}(\theta_{k,\ell}^{(i)}, \lambda_1^{(i)}, \lambda_2^{(i)}), \quad (13)$$

here the input to the model is the channel, the output is the precoder $\mathbf{F}_{k,\ell}^{(i+1)}$. That is, $\mathbf{F}_{k,\ell}^{(i)} = f(\mathbf{h}^{(i)}; \theta_{k,\ell}^{(i)})$, with $f(\cdot)$ denoting the output of the neural network parameterized by $\theta_{k,\ell}^{(i)}$, with input $\mathbf{h}^{(i)}$ (the channel at the i th iteration). The step sizes $\eta_{\theta_{k,\ell}}^{(i)}$, $\eta_{\lambda_1}^{(i)}$, and $\eta_{\lambda_2}^{(i)}$ may be obtained using backtracking line search, exponential decaying sequence, or something else that decays as the iteration index i increases. Reference [9] and reference therein for more information about what choice of neural network you should use.

Hence, the algorithm first finds $\mathbf{F}_{k,\ell}^{(i+1)}$ by solving (13) by fixing λ_1 and λ_2 to be $\lambda_1^{(i)}$ and $\lambda_2^{(i)}$, respectively. After obtaining $\mathbf{F}_{k,\ell}^{(i+1)}$, then $\lambda_1^{(i+1)}$ will be obtained using (12b). Finally, $\lambda_2^{(i+1)}$ will be obtained using (12c). Since it is $\frac{1}{L_k} \sum_{\ell=1}^{L_k} R^*(T_k, \mathbf{F}_{k,\ell})$ that needs to be maximized, variables need to be solved for all $\ell = 1, \dots, L_k$ and $k = 1, \dots, K$.

K. Reformulating in AWGN to simplify channel dispersion

Since precoder $\mathbf{F}$ was not included in Rayleigh Fading channel dispersion equation $V_{iid}(P)$ we will consider an AWGN channel for simplicity.

$$\mathbf{y} = \sum_{k=1}^K \mathbf{F}_k \mathbf{s}_k + \mathbf{w} \in \mathbb{C}^{N_R},$$

where $\mathbf{F}_k \in \mathbb{C}^{N_k \times d_k}$ denotes the precoding matrix for the k th user, d_k denotes the number of spatial streams user k is sending to the AP (and $d_k \leq N_k$), $\mathbf{s}_k \in \mathbb{C}^{d_k}$ denotes the transmitted signal for the k th user, $\mathbf{w} \in \mathbb{C}^{N_R}$ denotes the AWGN, with N_k denoting the number of antennas at the k th user and N_R denotes the number of antennas at the AP. It is assumed that $E[\mathbf{s}_k \mathbf{s}_k^H] = \mathbf{I}_{N_k}$.

The optimization problem then becomes,

$$\begin{aligned} \min_{\mathbf{F}_{k,\ell}, n_k, \epsilon_k} \quad & \sum_{k \in K} w_k \frac{n_k}{f_s} \\ \text{s.t.} \quad & \frac{B_k}{n_k} \leq \max_{\mathbf{F}_{k,\ell}} \frac{1}{L_k} \sum_{\ell=1}^{L_k} R^*(T_k, \mathbf{F}_{k,\ell}), \quad \forall k \in \mathcal{K} \\ & n_k \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K} \\ & \|\mathbf{F}_{k,\ell}\|_F^2 \leq P_{k,\ell}, \quad \forall \ell \in [0, L_k], \forall k \in \mathcal{K}. \end{aligned}$$

where $R^*(T_k, \mathbf{F}_{k,\ell})$ is,

$$R^*(T_k, \mathbf{F}_{k,\ell}) = C_{k,\ell}(\mathbf{F}_{k,\ell}) - \sqrt{\frac{V_k(\mathbf{F}_{k,\ell})}{T_{k,\ell}}} Q^{-1}(\epsilon_k), \quad \forall \ell \in [0, L_k] \forall k \in \mathcal{K}$$

The Capacity and Channel Dispersion equation can be for AWGN MIMO can be derived from [10] which presents the equations with SNR ρ .

$$\begin{aligned} \mathbf{A}_{k,\ell}(\mathbf{F}_{k,\ell}) &= \sigma^{-2} \mathbf{H}_{k,\ell} \mathbf{F}_{k,\ell} \mathbf{F}_{k,\ell}^H \mathbf{H}_{k,\ell}^H, \\ C_{k,\ell}(\mathbf{F}_{k,\ell}) &= \log_2 \det \left(\mathbf{I} + \mathbf{A}_{k,\ell}(\mathbf{F}_{k,\ell}) \right), \\ V_{k,\ell}(\mathbf{F}_{k,\ell}) &= \sum_i \frac{\lambda_i (\lambda_i + 2)}{(\lambda_i + 1)^2} (\log_2 e)^2 \\ &= \text{tr} \left[\mathbf{A}_{k,\ell} (\mathbf{A}_{k,\ell} + 2\mathbf{I}) (\mathbf{A}_{k,\ell} + \mathbf{I})^{-2} \right] (\log_2 e)^2, \end{aligned}$$

To solve this optimization problem the earlier method of solving for the Lagrangian can be used, where the precoder $\mathbf{F}_{k,\ell}$ can be parameterized by the neural network weights $\theta_{k,\ell}$ and gradient ascent be performed on the Lagrangian to obtain optimal precoder given the slack constraints.

L. Different formulations of R^* and solution

1. Per-Block Rate

Each coherence block can be treated as an independent transmission. The achievable rate can be optimized per block:

$$R_\ell^*(n_\ell, \mathbf{F}_\ell, \mathbf{H}_\ell) \approx C(\mathbf{F}_\ell, \mathbf{H}_\ell) - \sqrt{\frac{V(\mathbf{F}_\ell, \mathbf{H}_\ell)}{n_\ell}} Q^{-1}(\epsilon_\ell)$$

where n_ℓ is the blocklength of block ℓ and ϵ_ℓ is the target error probability for that block. This method allows per-block latency and optimization.

Updated Optimization Problem

$$\begin{aligned} \min_{\mathbf{F}_{k,\ell}, n_{k,\ell}, \epsilon_k} \quad & \sum_{k \in K} w_k \frac{n_k}{f_s} \\ \text{s.t.} \quad & \frac{B_{k,\ell}}{n_{k,\ell}} \leq \max_{\mathbf{F}_{k,\ell}} R_\ell^*(n_{k,\ell}, \mathbf{F}_{k,\ell}, \mathbf{H}_\ell), \quad \forall k \in \mathcal{K} \\ & n_{k,\ell} \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K} \\ & \|\mathbf{F}_{k,\ell}\|_F^2 \leq P_{k,\ell}, \quad \forall \ell \in [0, L_k], \forall k \in \mathcal{K}. \end{aligned}$$

Updating $B_{k,\ell}$ post optimization of $n_{k,\ell}$

Once $n_{k,\ell}$ is minimized through the alternating optimization process.

If

$$\frac{B_{k,\ell}}{n_{k,\ell}} > R_\ell^*(n_{k,\ell}, \mathbf{F}_{k,\ell})$$

Update and reduce $B_{k,\ell}$ to $B_{k,\ell} = \lfloor n_{k,\ell} \times R_\ell^*(n_{k,\ell}, \mathbf{F}_{k,\ell}) \rfloor$

Asynchronality Optimization

Once blocklength optimization process is over, we would obtain a list of all possible n_k values for user- k in the interval $[n_{\min}, n_{\max}]$.

To solve the asynchronality we can search through the possible n_k values for user- k received from the previous alternating-optimization process to find the correct n_k values for each user k to minimize the following objectives,

1. Weighted Average Latency

We can use the weighted average latency, where the weights are decided by $w_k = \frac{1}{CNR_k}$ ($\frac{1}{SNR_k}$ in the case of AWGN).

$$\min_{\mathbf{F}_{k,\ell}, n_{k,\ell}, \epsilon_k} \sum_{k \in K} w_k \frac{n_k}{f_s}$$

However this might just end up reducing the overall latency of the system instead of reducing the latency as there is no tradeoff(of resources) that occurs between the users to justify the use of weights.

2. Max-Min

To create a more fair asynchronous optimization we use max-min optimization.

$$\max_{\mathbf{F}_{k,\ell}, n_k, \epsilon_k} \min_{k \in K} \sum_{k \in K} w_k \frac{n_k}{f_s}$$

This would ensure that the latency of the system is the same as the slowest user rendering the asynchronality to be equal to 0.

2. Average Rate over Coherence Blocks

In the standard block-fading FBL model, a codeword spans multiple coherence blocks, each with independent channel realizations. The achievable rate is approximated using the expectation of capacity and dispersion over coherent blocks:

Assume there are L coherent blocks, and each block has n_ℓ symbols.

$$R_k^*(n_\ell, \mathbf{F}_\ell, \mathbf{H}_\ell) = \frac{1}{n} \sum_{\ell=1}^L n_\ell C(\mathbf{F}_\ell, \mathbf{H}_\ell) - \sqrt{\frac{\frac{1}{n} \sum_{\ell=1}^L n_\ell V(n_\ell, \mathbf{F}_\ell, \mathbf{H}_\ell)}{n}} Q^{-1}(\epsilon)$$

where $n = \sum_{\ell} n_\ell$ (total blocklength), $C(h_\ell)$ is the Shannon capacity of block ℓ , $V(h_\ell)$ is the channel dispersion, and ϵ is the target codeword error probability.

I think this allows joint decoding while the other method doesnt.

M. Simulation of Independent Sub-block Transmission

Experiment Setup

We consider a block-fading channel where the channel is constant over a coherence block and changes independently across different coherence blocks.

Variable	Definition
K	Number of Users
L	Number of sub-blocks (coherence blocks)
ℓ	Sub-block index
T	Coherence blocklength (symbols per sub-block)
N_r	Number of Receive Antennas
N_t	Number of Transmit Antennas
f_s	Symbol rate
$n_{k,\ell}$	Blocklength for user k at sub-block ℓ
ϵ_k	Target error probability for user k
$\mathbf{H} \in \mathbb{C}^{K \times L \times N_r \times N_t}$	Channel matrix
$\mathbf{F} \in \mathbb{C}^{K \times L \times N_t \times d_k}$	Precoder matrix
$\mathbf{X} \in \mathbb{C}^{K \times L \times d_k \times n_l}$	Transmitted signal
$B_{k,\ell}$	Number of bits transmitted for user k in sub-block ℓ
$R_{k,\ell}^*(\mathbf{H}_\ell, \mathbf{F}_\ell, n_\ell)$	Finite-blocklength rate for user k in sub-block ℓ
$C_{k,\ell}(\mathbf{H}_\ell, \mathbf{F}_\ell, n_\ell)$	Shannon capacity of channel H_ℓ
$V_{k,\ell}(\mathbf{H}_\ell, \mathbf{F}_\ell, n_\ell)$	Channel dispersion of channel H_ℓ
P_k	Transmit power constraint of user k
M_ℓ	Modulation order in sub-block ℓ
γ_k	Signal-to-Interference-plus-Noise Ratio (SINR)

TABLE I: Table of Variables and Definitions

Aim

The experiment aims to reduce the number of symbols transmitted in each sub-block to minimize the latency of the system. This is done by optimizing

$$\begin{aligned}
& \min_{\mathbf{F}_{k,\ell}, n_{k,\ell}, \epsilon_k} \sum_{k \in K} w_k \frac{n_k}{f_s} \\
& \text{s.t. } \frac{B_{k,\ell}}{n_{k,\ell}} \leq \max_{\mathbf{F}_{k,\ell}} R_\ell^*(n_{k,\ell}, \mathbf{F}_{k,\ell}, \mathbf{H}_{k,\ell}), \quad \forall k \in \mathcal{K} \\
& n_{k,\ell} \in [n_{\min}, n_{\max}], \quad \forall k \in \mathcal{K} \\
& \|\mathbf{F}_{k,\ell}\|_F^2 \leq P_{k,\ell}, \quad \forall \ell \in [0, L_k], \forall k \in \mathcal{K}.
\end{aligned}$$

(The optimization problem has been rewritten from previous formulations to optimize per sub-block by considering each sub-block as an independent transmission)

The Capacity and Channel Dispersion equation for AWGN MIMO can be derived from [10] which presents the equations as

$$\begin{aligned}
\mathbf{A}_{k,\ell}(\mathbf{H}_{k,\ell}, \mathbf{F}_{k,\ell}) &= \sigma^{-2} \mathbf{H}_{k,\ell} \mathbf{F}_{k,\ell} \mathbf{F}_{k,\ell}^H \mathbf{H}_{k,\ell}^H, \\
C_{k,\ell}(\mathbf{H}_{k,\ell}, \mathbf{F}_{k,\ell}) &= \log_2 \det (\mathbf{I} + \mathbf{A}_{k,\ell}(\mathbf{H}_{k,\ell}, \mathbf{F}_{k,\ell})), \\
V_{k,\ell}(\mathbf{H}_{k,\ell}, \mathbf{F}_{k,\ell}) &= (\log_2 e)^2 \sum_i \frac{\lambda_i(\lambda_i + 2)}{(\lambda_i + 1)^2},
\end{aligned} \tag{14}$$

where λ_i are the eigenvalues of $\mathbf{A}_{k,\ell}(\mathbf{F}_{k,\ell})$.

$$R_{k,\ell}^*(n_{k,\ell}, \mathbf{F}_{k,\ell}, \mathbf{H}_{k,\ell}) \approx C(\mathbf{F}_{k,\ell}, \mathbf{H}_{k,\ell}) - \sqrt{\frac{V(\mathbf{F}_{k,\ell}, \mathbf{H}_{k,\ell})}{n_{k,\ell}}} Q^{-1}(\epsilon_k)$$

Training Algorithm

SubBlocklength-Precoder Optimization for Training

Ensure: Feasible solutions $\{(n_{k,\ell}^*, \mathbf{F}_{k,\ell}^*, R_{k,\ell}^*)\}$

- 1: **for** $k = 1$ **to** K **do** ▷ Loop over users
- 2: Normalize channel: $\mathbf{H}_k \leftarrow (\mathbf{H}_k - \text{mean}(\mathbf{H}_k)) / \sqrt{\text{var}(\mathbf{H}_k)}$
- 3: **for** $\ell = 1$ **to** L **do** ▷ Loop over sub-blocks
- 4: $\mathcal{S}_{k,\ell} \leftarrow \text{OPTIMIZE_SUBBLOCKLENGTH_PRECODER}(f_\theta, \mathbf{F}_{\text{init}}, n_{\ell,\max}, n_{\ell,\min}, \Delta n_\ell, \lambda_{\text{rate}}^{(0)}, \lambda_{\text{power}}^{(0)})$
- 5: Select feasible tuple $(n_{k,\ell}^*, \mathbf{F}_{k,\ell}^*, R_{k,\ell}^*) = \mathcal{S}_{k,\ell}[-1]$
- 6: **end for**
- 7: **end for**
- 8: **return** all feasible solutions

Algorithm 1: OPTIMIZE SUBBLOCKLENGTH-PRECODER**Ensure:** Feasible tuples $\{(n_\ell^*, \mathbf{F}_\ell^*, R_{\text{FBL}}^*, \lambda_{\text{rate}}^*, \lambda_{\text{power}}^*)\}$

```

1: Initialize PrecoderNet  $f_\theta$  and Adam optimizer with learning rate  $\eta_{\text{net}}$ 
2:  $\lambda_{\text{rate}} \leftarrow \lambda_{\text{rate}}^{(0)}$ 
3:  $\lambda_{\text{power}} \leftarrow \lambda_{\text{power}}^{(0)}$ 
4:  $Results \leftarrow \emptyset$ 
5: for  $n_\ell = n_{\ell, \max}$  to  $n_{\ell, \min}$  step  $-\Delta n_\ell$  do
6:   if  $n_\ell = n_{\ell, \max}$  then  $\triangleright$  (It is observed that the program takes a larger number of
     epochs to find the initial precoder)
7:      $epochs \leftarrow 2000$ 
8:   else
9:      $epochs \leftarrow epochs_{\text{per-}n_\ell}$ 
10:  end if
                                      $\triangleright$  Optimize precoder for current sub-blocklength
11:   $(\mathbf{F}^*, \lambda'_{\text{rate}}, \lambda'_{\text{power}}, R_{\text{FBL}}, c_{\text{rate}}, c_{\text{power}}, \mathcal{L}) \leftarrow \text{OPTIMIZEPRECODER}(f_\theta, \mathbf{F}_{\text{init}}, n_\ell, epochs, \lambda_{\text{rate}}, \lambda_{\text{power}})$ 
12:   $\lambda_{\text{rate}} \leftarrow \lambda'_{\text{rate}}$ 
13:   $\lambda_{\text{power}} \leftarrow \lambda'_{\text{power}}$ 
14:  if  $c_{\text{rate}} \leq 0$  and  $c_{\text{power}} \leq 0$  then
                                      $\triangleright$  Rate and Precoder constraints are satisfied for the given sub-blocklength
15:     $Results \leftarrow Results \cup \{(n_\ell, \mathbf{F}^*, R_{\text{FBL}}, \lambda_{\text{rate}}, \lambda_{\text{power}})\}$ 
16:  end if
17: end for
18: return  $Results$ 

```

ALGORITHM 2: OPTIMIZEPRECODER

Require: Precoder network f_θ , initial precoder $\mathbf{F}$, number of epochs E , Lagrange multipliers

$\lambda_{rate}, \lambda_{power}$, learning rates $(\eta_{net}, \eta_{rate}, \eta_{power})$

Ensure: Optimized precoder $\mathbf{F}^*$, updated multipliers $(\lambda_{rate}, \lambda_{power})$, achievable rate R_{FBL} ,

constraint violations (c_{rate}, c_{power})

- 1: $Losses \leftarrow \emptyset$
- 2: **for** $e = 1$ **to** E **do**
- 3: $\mathbf{F}_{out} \leftarrow f_\theta(\mathbf{H}_{kl})$
- 4: Zero optimizer gradients
- 5: $(\mathcal{L}, R_{\text{FBL}}, P_F, c_{rate}, c_{power}) \leftarrow \text{LAGRANGIANLOSS}(\mathbf{F}_{out}, \lambda_{rate}, \lambda_{power})$
 $\triangleright$ Early Stopping: If rate and power constraints are satisfied for the given
 blocklength and optimization runs for min epochs, break the loop
- 6: **if** $c_{rate} \leq 0$ **and** $c_{power} \leq 0$ **and** $e \geq 10$ **then**
- 7: **break**
- 8: **end if**
- 9: $(\lambda_{rate}, \lambda_{power}) \leftarrow \text{UPDATELAMBDAS}(\lambda_{rate}, \lambda_{power}, c_{rate}, c_{power}, \eta_{rate}, \eta_{power})$
- 10: Backpropagate $\mathcal{L}$
- 11: Optimizer step
- 12: Append $\mathcal{L}$ to $Losses$
- 13: **end for**
- 14: $\mathbf{F}^* \leftarrow f_\theta(\mathbf{H}_{kl})$
- 15: Project $\mathbf{F}^*$ to valid precoder form
- 16: **return** $\mathbf{F}^*$, optimizer, $\lambda_{rate}, \lambda_{power}, R_{\text{FBL}}, P_F, c_{rate}, c_{power}, Losses$

Algorithm 3: LAGRANGIANLOSS**Require:** Precoder $\mathbf{F}_{\mathbf{k},\ell}$, dual variables $(\lambda_{rate}, \lambda_{power})$,1: System parameters $(\mathbf{H}_{k\ell}, n_{k,\ell}, \sigma^2, \epsilon_k, B_{k,l}, P_k)$ **Ensure:** Lagrangian loss $\mathcal{L}$, rate R_{FBL} , power $\|\mathbf{F}_{\mathbf{k},\ell}\|_F^2$,2: Constraint violations (c_{rate}, c_{power}) 3: $\mathbf{A}_{k,\ell} \leftarrow \sigma^{-2} \mathbf{H}_{k\ell} \mathbf{F}_{\mathbf{k},\ell} \mathbf{F}_{\mathbf{k},\ell}^H \mathbf{H}_{k\ell}^H$

▷ Compute finite-blocklength capacity

4: $C_{k,\ell} \leftarrow \log_2 \det(\mathbf{I} + \mathbf{A}_{k,\ell})$

▷ Compute channel dispersion

5: Compute eigenvalues $\{\lambda_i\}$ of $\mathbf{A}_{\mathbf{k},\ell}$ 6: $V_{k,\ell} \leftarrow \sum_i \frac{\lambda_i(\lambda_i+2)}{(\lambda_i+1)^2} (\log_2 e)^2$

▷ Finite-blocklength achievable rate

7: $R_{\text{FBL}} \leftarrow C - \sqrt{\frac{V}{n_{k,\ell}}} Q^{-1}(\epsilon)$

▷ Evaluate constraints

8: $F_{power} \leftarrow \|\mathbf{F}_{\mathbf{k},\ell}\|_F^2$ 9: $c_{rate} \leftarrow \frac{B_{k,\ell}}{n_{k,\ell}} - R_{\text{FBL}}$ 10: $c_{power} \leftarrow F_{power} - P_k$

▷ Soft constraint handling (penalty only if violated)

▷ LeakyReLU is

used as it was observed that the algorithm does not achieve maximum rate or precoder power and plays safe to satisfy the constraints. Using LeakyReLU incentivizes a higher rate and power while giving a larger penalty to breaking them.

11: $c_{rate} \leftarrow \text{LeakyReLU}(c_{rate})$ 12: $c_{power} \leftarrow \text{LeakyReLU}(c_{power})$

▷ Primal Dual Lagrangian objective

13: $\mathcal{L} \leftarrow -R_{\text{FBL}} + \lambda_{rate} \cdot c_{rate} + \lambda_{power} \cdot c_{power}$ 14: **return** $(\mathcal{L}, R_{\text{FBL}}, F_{power}, c_{rate}, c_{power})$

Algorithm 4: UPDATELAMBDA (Primal–Dual Update)**Require:** Current dual variables $(\lambda_{rate}, \lambda_{power})$,1: Constraint violations c_{rate} (rate), c_{power} (power),2: Dual step sizes $(\eta_{rate}, \eta_{power})$ **Ensure:** Updated dual variables $(\lambda_{rate}, \lambda_{power})$

▷ This procedure performs a updation step in the primal–dual optimization

▷ Update Lagrange multipliers

3: $\lambda_{rate} \leftarrow \lambda_{rate} + \eta_{rate} \cdot c_{rate}$ 4: $\lambda_{power} \leftarrow \lambda_{power} + \eta_{power} \cdot c_{power}$ 5: $\lambda_{rate} \leftarrow \max(0, \lambda_{rate})$ 6: $\lambda_{power} \leftarrow \max(0, \lambda_{power})$ 7: **return** $(\lambda_{rate}, \lambda_{power})$

Precoder Architecture

UplinkMLP PrecoderNet Architecture

- **Parameters:**

- N_r : Number of receive antennas
- N_t : Number of transmit antennas
- d_k : Number of data streams per user
- Input dimension: $in_dim = 2 \cdot N_r \cdot N_t$ (concatenated real and imaginary parts)
- Output dimension: $out_dim = 2 \cdot N_t \cdot d_k$
- Hidden dimension: $hidden = 4 \cdot (4 \cdot out_dim)$

- **Input:** Flattened channel matrix $H_{k\ell} \in \mathbb{C}^{N_r \times N_t}$

- Real and imaginary parts concatenated: $\text{vec}([H_{k\ell}]_{\text{real}}, [H_{k\ell}]_{\text{imag}}) \in \mathbb{R}^{in_dim}$

- **Fully Connected Layers:**

- 1) Linear($in_dim \rightarrow hidden$), ReLU
- 2) Linear($hidden \rightarrow hidden/2$), ReLU
- 3) Linear($hidden/2 \rightarrow hidden/4$), ReLU
- 4) Linear($hidden/4 \rightarrow out_dim$)

- **Output:** Precoder vector $F_{\text{out}} \in \mathbb{R}^{out_dim}$, representing real and imaginary parts of the precoder for the sub-block.

Layer	Input $\rightarrow$ Output	Activation
Input	$in_dim = 2N_r N_t$	-
Linear 1	$in_dim \rightarrow hidden = 4 \cdot (4 \cdot out_dim)$	ReLU
Linear 2	$hidden \rightarrow hidden/2$	ReLU
Linear 3	$hidden/2 \rightarrow hidden/4$	ReLU
Linear 4	$hidden/4 \rightarrow out_dim = 2N_t d_k$	-
Output	$out_dim = 2N_t d_k$	-

Testing Algorithm

Sub-Blocklength per User Testing

Require: Number of users K , simulation parameters $(n_{\ell,\max}, n_{\ell,\min}, \Delta n_{\ell})$, Trained precoders

$\{\mathbf{F}_{k,\ell}\}$, channel statistics (μ_k, σ_k^2) for all k from training

Ensure: Rate histories $\{\mathcal{R}_k\}_{k=1}^K$, blocklength histories $\{\mathcal{N}_k\}_{k=1}^K$, Bits per Symbol histories

$\{\mathcal{B}_k\}_{k=1}^K$

```

1: for  $k = 1$  to  $K$  do
2:    $\mathbf{H}_k \leftarrow (\mathbf{H}_k - \mu_k) / \sqrt{\sigma_k^2}$ 
3:   Initialize:  $\mathcal{R}_k \leftarrow \emptyset, \mathcal{N}_k \leftarrow \emptyset, \mathcal{B}_k \leftarrow \emptyset$ 
4:   for  $\ell = 1$  to  $L_k$  do
5:     Initialize histories:  $\mathcal{R}_{k,\ell} \leftarrow \emptyset, \mathcal{N}_{k,\ell} \leftarrow \emptyset, \mathcal{B}_{k,\ell} \leftarrow \emptyset$ 
6:     Load trained precoder  $\mathbf{F}_{k,\ell}$ 
7:     for  $n_{\ell} = n_{\ell,\max}$  to  $n_{\ell,\min}$  step  $-\Delta n_{\ell}$  do
8:       Set blocklength  $n_{\ell}$  for user  $k$ , block  $\ell$ 
9:       Compute achievable rate  $R_{\text{FBL}}^{k,\ell}$ 
10:      Compute required rate per symbol  $B_k/n_{\ell}$ 
11:      if  $\frac{B_k}{n_{\ell}} > R_{\text{FBL}}^{k,\ell}$  then
           $\triangleright$  Constraint violated: minimum feasible blocklength reached
12:        Restore blocklength to  $n_{\ell} + \Delta n_{\ell}$ 
13:        break
14:      end if
15:      Append  $R_{\text{FBL}}^{k,\ell}$  to  $\mathcal{R}_{k,\ell}$ 
16:      Append  $n_{\ell}$  to  $\mathcal{N}_{k,\ell}$ 
17:      Append  $B_k/n_{\ell}$  to  $\mathcal{B}_{k,\ell}$ 
18:    end for
19:    Append  $\mathcal{R}_{k,\ell}$  to  $\mathcal{R}_k$ 
20:    Append  $\mathcal{N}_{k,\ell}$  to  $\mathcal{N}_k$ 
21:    Append  $\mathcal{B}_{k,\ell}$  to  $\mathcal{B}_k$ 
22:  end for
23: end for
24: return  $(\{\mathcal{R}_k\}, \{\mathcal{B}_k\}, \{\mathcal{N}_k\})$ 

```

Problems with this approach

The problems with this approach stem from how we initialize the simulation.

An example initial parameters for 2 identical users used in the current simulation is as follows:

Parameter	User 1	User 2
Number of users K	2	
Receive antennas N_r	16	16
Transmit antennas N_t	4	4
Number of sub-blocks L_k	5	5
Bits per sub-block $B_{k,l}$	500	500
Symbols per sub-block(coherent block) T_k	250	250
Initial sub-block length $n_{k,\ell}$	250	250
SNR (dB)	5.0	5.0
Transmit power P_t	20.0	20.0
Symbol frequency f_s (Hz)	15,000	15,000
Target error probability ϵ	10^{-10}	10^{-10}
Training seed	0	
Testing seed	1	

The initial parameters assume the number of sub-blocks L , this is a problem as the number of sub-blocks is derived from the equation $L = N/T$, where T (coherence time) is the only channel constant.

Thus it is not right to set the initial value of L as it is a derived quantity.

As the current algorithm reduces the sub-blocklength n_l , this could create a scenario post-optimization where due to reducing n_l and having constant T , sub-block l has less symbols and could also contain the symbols of the next sub-block.

For example assume,

$$T = 200, n_1 = 200, n_2 = 200, L = 2,$$

post optimization we optimize n_1 and n_2 to become,

$$T = 200, n_1 = 100, n_2 = 80$$

Thus in this scenario sub-block 1 can hold symbols of sub-block 1 and sub-block 2.

This is because $n_1(100) + n_2(80) < T(200)$, thus all the symbols can fit in the first coherent block.

Proposed Solution

Lets remove initial values of L_k and $n_{k,\ell}$ and rewrite the Initial Parameters to only contain Total Bits B_k and coherance time T_k .

Thus the rewritten initial parameters are as follows:

Parameter	User 1	User 2
Number of users K	2	
Receive antennas N_r	16	16
Transmit antennas N_t	4	4
Total Bits per user B_k	5000	5000
Symbols per sub-block(coherent block) T_k	250	250
SNR (dB)	5.0	5,0
Transmit power P_t	20.0	20.0
Symbol frequency f_s (Hz)	15,000	15,000
Target error probability ϵ	10^{-10}	10^{-10}
Training seed	0	
Testing seed	1	

We will try to solve the problem of changing L_k to a derived quantity by assuming initial value of $L_k = 1$ and $n_{k,1} = T_k$.

This means initially all of user bits B_k is encoded into T_k symbols($n_{k,1} = T_k$) within a single sub-block(coherent block)($L_k = 1$). Thus initially the simulation has a high bit/symbol ratio of $B_k/n_{k,1} = B_k/T_k$ which corresponds to a higher modulation order of $M_{k,\ell} = 2^{B_k/T_k}$.

However this might not satisfy the rate and power constraints, then we try to optimize L_k and $B_{k,\ell}$ to reduce the sub-blocklength($n_{k,\ell}$) to satisfy the constraints.

– If the sub-blocklength $n_{k,\ell} = T_k$ does not satisfy the rate and power constraints, then we reduce $B_{k,\ell}$ in the range $[B_k, 1]$ until the constraints are satisfied. Then B_k is then updated to $B_k = B_k - B_{k,\ell}$

– If remaining $B_k > 0$, then we increase $L_k = L_k + 1$ and repeat the process on the next sub-block until all bits are encoded into sub-blocks ie: remaining $B_k = 0$.

Thus instead of fixing the number of sub-blocks and blocklengths a priori, we derive them dynamically from the total remaining bits and coherence time. Starting from a T_k -symbol transmission, the sub-blocklength is progressively decreased if the rate and power constraints are satisfied. If reduction is not possible, the transmitted bits are split across multiple sub-blocks.

I believe this is also optimal as having less sub-blocks with its high sub-blocklength (ie: having sub-blocklength($n_{k,\ell}$) = coherence interval (T_k)) result in lower latency in the current simulation as compared to more sub-blocks with lower sub-blocklengths. I do not have a proof for this but I believe an intuitive explanation is the fact that having a lower $n_{k,\ell}$ results in a larger **dispersion penalty** thus reducing the overall rate. This results in a higher total n_k and thus higher latency when you sum the smaller $n_{k,\ell}$ across more sub-blocks. Thus it is better to have less sub-blocks with higher $n_{k,\ell}$ than more sub-blocks with lower $n_{k,\ell}$.

The refined algorithm is as follows:

Algorithm: Dynamic SubBlocklength-Precoder Optimization for Training

Require: User payloads $\{B_k\}_{k=1}^K$, coherence blocklengths $\{T_k\}_{k=1}^K$,

- 1: Channels $\{\mathbf{H}_k\}_{k=1}^K$,
- 2: Initial dual variables $(\lambda_{rate}^{(0)}, \lambda_{power}^{(0)})$,
- 3: Sub-block parameters $(n_{\ell, \min}, \Delta n_\ell)$

Ensure: Number of sub-blocks $\{L_k\}$, sub-blocklengths $\{n_{k,\ell}^*\}$, precoders $\{\mathbf{F}_{k,\ell}^*\}$, rates $\{R_{k,\ell}^*\}$

- 4: **for** $k = 1$ **to** K **do** ▷ Loop over users
- 5: Normalize channel: $\mathbf{H}_k \leftarrow (\mathbf{H}_k - \text{mean}(\mathbf{H}_k)) / \sqrt{\text{var}(\mathbf{H}_k)}$
- 6: $B_k^{\text{rem}} \leftarrow B_k$
- 7: $L_k \leftarrow 1$
- 8: $\ell \leftarrow 0$
- 9: $n_{\ell, \max} \leftarrow T_k$
- 10: **while** $B_k^{\text{rem}} > 0$ **do**
- 11: $\mathcal{S}_{k,\ell} \leftarrow \text{OPTIMIZE_SUBBLOCKLENGTH_PRECODER}(f_\theta, \mathbf{F}_{\text{init}}, n_{\ell, \max}, n_{\ell, \min}, \Delta n_\ell, \lambda_{rate}^{(0)}, \lambda_{power}^{(0)})$
- 12: $(n_{k,\ell}^*, \mathbf{F}_{k,\ell}^*, R_{k,\ell}^*) \leftarrow \mathcal{S}_{k,\ell}[-1]$ ▷ Select smallest feasible blocklength
- 13: $B_{k,\ell} \leftarrow \min(B_k^{\text{rem}}, \lfloor n_{k,\ell}^* \cdot R_{k,\ell}^* \rfloor)$ ▷ Determine transmitted bits in this sub-block
- 14: $B_k^{\text{rem}} \leftarrow B_k^{\text{rem}} - B_{k,\ell}$
- 15: **if** $B_k^{\text{rem}} > 0$ **then**
- 16: $L_k \leftarrow L_k + 1$
- 17: $\ell \leftarrow \ell + 1$
- 18: **end if**
- 19: **end while**
- 20: **end for**
- 21: **return** $(\{L_k\}, \{n_{k,\ell}^*\}, \{\mathbf{F}_{k,\ell}^*\}, \{R_{k,\ell}^*\})$

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